

Article

Integrating Reading, Writing, and Digital Tools in Science: A Participatory-Design Study of the InSPECT Framework

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Abstract

The purpose of this study was to engage high school science teachers as co-design partners in refining and extending instructional frameworks to support multiple-document reading and writing in science classrooms. Using a participatory mixed-methods design, the project adapted the InSPECT framework for secondary science, developed professional development (PD) materials to introduce the framework, and explored the role of generative artificial intelligence (AI) in lesson planning. Five virtual focus group sessions guided the co-design of PD activities, followed by a pilot implementation in one biology classroom. Data included focus group and interview transcripts, surveys, and student work artifacts. Analyses examined teachers' perceptions of PD features, framework usability, and student engagement. Teachers valued PD that was practical, relevant, and feasible within classroom constraints and described the frameworks as clear, stepwise structures that supported lesson design and literacy integration. Student work showed that paraphrasing was an accessible entry point, while bridging, elaboration, and source evaluation required additional modeling. Teachers viewed generative AI as a promising planning aid but expressed concerns about accuracy and ethics. Findings informed revisions emphasizing discipline-specific exemplars, scaffolds for higher-order strategies, and AI-literacy modules, illustrating how participatory design can yield feasible, teacher-centered PD.

Keywords: disciplinary literacy; science education; multiple-document comprehension; professional development; participatory design; generative artificial intelligence (AI)



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1. Introduction

In today's information-rich world, people are expected to integrate information from multiple sources, evaluate its credibility, and communicate their understanding in writing—skills emphasized in international assessments ([Organisation for Economic Co-operation and Development, 2019](#); [Barzilai et al., 2018](#); [Tal-Savir et al., 2021](#)) and in disciplinary standards such as the Next Generation Science Standards (NGSS; [National Research Council, 2013](#); [Pruitt, 2014](#)). To meet the complex demands of reading, writing, and reasoning with evidence in science, students need disciplinary literacy instruction tailored to science classrooms ([Goldman et al., 2016](#); [Hubbard, 2021](#); [Shanahan & Shanahan, 2008](#)). Yet, national survey data reveal that high school science teachers devote little time to instruction that supports these forms of synthesis and evidence-based writing ([Drew & Thomas, 2017](#); [Drew et al., 2017](#)). This mismatch underscores the need for research that

develops practical, teacher-centered approaches to embedding integrated reading–writing activities in science instruction (Blake & Traga Philippakos, 2025; Salerno et al., 2020).

The present study responds to this need by engaging high school science teachers as co-design partners in developing classroom-ready approaches to support multiple-document integrated reading and writing. Through focus groups and a pilot implementation, teachers evaluated professional development activities, revised instructional materials, and reflected on the feasibility of embedding these practices into their science curricula. The instructional materials developed in this work included practical approaches to the use of generative AI as a tool for lesson design, providing teachers with new ways to create and adapt tasks that integrate evidence-based literacy strategies with their classroom content. In doing so, this work demonstrates how co-design and AI integration can help bridge the research-to-practice gap and make disciplinary literacy goals more actionable in everyday instruction.

1.1. Theoretical Foundations: Multiple-Document Integrated Reading and Writing

Research in multiple-document comprehension is largely driven by theories of text and discourse comprehension which assume that successful reading emerges when readers construct an elaborated and coherent mental model of the text information (McNamara & Magliano, 2009). Many early theories focused on how readers made sense of texts in isolation. However, as reading in digital environments became more accessible and the sources and quantities of available information increased, these theories were expanded to consider how readers would need to organize, evaluate, synthesize, and make sense of information derived from multiple sources. While a full review of the different theories of multiple-document comprehension is outside the scope of this manuscript, it is important to note that all of these theories highlight the likelihood that readers will encounter content of varying quality, credibility, and complexity across texts that introduces both complementary and conflicting information between them (see List & Alexander, 2019).

Early theoretical models, such as the Documents Model (Perfetti et al., 1999) emphasized that, in order to make connections and resolve inconsistencies, readers need to effectively evaluate the sources of the texts. They must construct both representations of the individual texts (i.e., intratext) as well as a representation of the larger space that includes how those texts are related to one another (i.e., intertext). This requires the reader to attend to and represent source information as well as the text content. Additional models built upon this foundation have also highlighted the need to consider both how students understand the task and monitor their progress towards their goals as well as how students create a product (often written) that externalizes their understanding of the topic (e.g., MD-TRACE, RESOLV; IF-MT, List & Alexander, 2019). This production requires students to not only have a strong mental representation of the texts, but also a variety of skills in writing and argumentation (e.g., Bråten et al., 2025; McCarthy et al., 2022; Sonia et al., 2022). These models suggest that successful multiple-document comprehension requires not only evaluating and integrating sources but also producing written representations of knowledge.

While multiple-document comprehension is relevant across disciplines, the design of multiple-document tasks should align with the discipline-specific practices that students are expected to develop. Following Goldman et al. (2016), disciplinary literacy emphasizes skills such as evaluating sources, reasoning with evidence, and constructing explanations consistent with the ways knowledge is generated and communicated within a discipline. By embedding these tasks within the specific practices of science, students are not only practicing multiple-document comprehension but also developing the domain-specific literacy skills necessary to reason and communicate like scientists.

1.2. Bridging Theory and Practice: Instructional Structures and Teacher Learning

While theoretical models of multiple-document comprehension highlight the skills students need to evaluate, integrate, and write from multiple sources, teachers require concrete instructional structures to translate these insights into practice. Text sets, and, in particular, quad text sets, have emerged as one such framework (W. E. Lewis & Strong, 2020; W. Lewis & Walpole, 2016; Lupo et al., 2018, 2019). Quad text sets are organized around four different types of texts, including one challenging on- or above-grade-level “target” text, such as a peer-reviewed scientific journal article or similar text, that may be challenging for students to comprehend without scaffolds. Supporting texts are sequenced to build background knowledge, introduce key vocabulary, and motivate engagement, while instructional routines guide students before, during, and after reading. In science classrooms, this structure has been used to integrate disciplinary literacy practices by scaffolding students with accessible videos or short articles to increasingly rigorous tasks such as vocabulary grouping, graphic organizers, debates, and written argumentation. The sequence culminates with students reading and evaluating a complex research article or other target text, positioning literacy not as an add-on but as part of authentic scientific inquiry.

At their core, quad text sets rest on several key premises that address the learning needs of high school science students (see W. E. Lewis & Strong, 2020 for a comprehensive review). First, students need regular opportunities to engage with complex disciplinary texts (Garderen et al., 2021; Lupo et al., 2019). Second, to comprehend these complex texts, students must learn and apply explicit strategies for evaluating sources and connecting ideas within and across those sources (e.g., Barzilai et al., 2018; Goldman & Scardamalia, 2013; McNamara, 2004, 2017). Third, students benefit from building background knowledge and motivation, which may be accomplished by engaging with varied sources such as informational articles, visual texts, or motivational pieces (Barzilai & Strømso, 2018; Hattan et al., 2024; List & Alexander, 2019; Lupo et al., 2019; Salerno et al., 2020).

One promising structure that helps teachers address these needs across each text in a quad text set is the Before–During–After (BDA) framework, which provides a clear sequence for integrating literacy instructional practices at each stage of reading (see W. E. Lewis & Strong, 2020). Before reading, teachers review strategies, such as evaluating the credibility of a source by examining its author, publisher, and purpose. During reading, students apply comprehension strategies such as generating self-explanations to monitor understanding and make connections across texts. After reading, students synthesize ideas through discussion and writing tasks. Applied to each of the four texts, the BDA framework helps students build background knowledge, connect ideas, and engage with increasingly complex disciplinary texts. Moreover, quad text sets are designed to be flexible and adaptable across disciplines (Freeman & Hardin, 2024), which is particularly important in high school science where instruction spans diverse domains such as biology, astronomy, and physical science. This adaptability, along with growing evidence from classroom-based research (e.g., Lupo et al., 2019), highlights quad text sets as a promising framework for operationalizing multiple-document comprehension in authentic science instruction.

Translating these instructional techniques and structure into classroom practice depends on teachers, who must design lessons, select appropriate texts, and establish the routines and procedures that guide students’ engagement. Research with pre-service teachers shows that while many view frameworks such as quad text sets as useful tools, they often focus narrowly on building background knowledge and may resist embedding literacy into science instruction, underscoring the importance of early preparation and support (Salerno et al., 2020). At the same time, studies with in-service teachers demonstrate that well-designed professional learning can meaningfully shift classroom practice. In a large-scale trial, intensive professional learning enabled high school biology

teachers to implement scaffolds and routines that emphasized reading and synthesizing information from multiple texts, with students outperforming peers on comprehension measures (Goldman et al., 2019). A year-long series by Lupo et al. (2023) similarly engaged secondary educators in using quad text sets and other disciplinary literacy practices, leading to more asset-based and knowledge-building approaches to intervention. Other design-based interventions illustrate that even brief cycles of PD and coaching can increase teachers' self-efficacy and expand the strategies they use to support student comprehension of complex science texts (Blake & Traga Philippakos, 2025; Thomas & Drew, 2021).

Beyond demonstrating that professional learning can influence classroom practice, recent work has examined how science teachers themselves perceive and evaluate PD focused on disciplinary literacy. Across studies, teachers emphasize that PD is most effective when it is collaborative, relevant to their curricular goals, and supported by explicit modeling of literacy strategies. A systematic review of disciplinary literacy PD found that successful programs typically balance explicit strategy instruction with opportunities for differentiation, reflection, and sustained application in classroom contexts (Howell et al., 2021). In a three-year design-based PD project, teachers reported that collaboration, adaptability, and opportunities to experience disciplinary literacy practices as learners before implementation were central to meaningful instructional change (Thomas & Drew, 2021). Additional work shows that PD integrating content knowledge and disciplinary literacy instruction enhances teachers' understanding and teaching effectiveness, particularly when supporting multilingual learners (Aguirre-Muñoz et al., 2024). Collectively, these findings suggest that teachers view PD as most impactful when it connects disciplinary goals with literacy instruction, provides space for co-design and adaptation, and aligns with teachers' classroom realities—principles that guided the participatory design approach in the present study.

1.3. Leveraging Generative AI to Support Instructional Development

Designing and implementing frameworks such as BDA requires considerable time and instructional planning. While these approaches can streamline lesson design in the long term, they often demand significant upfront preparation (W. E. Lewis & Strong, 2020). Such demands add to teachers' existing responsibilities and reflect broader challenges of sustaining instructional innovation in contexts where workload and time pressures are already high (National Education Association, 2022). These pressures are compounded by the complexities of integrating new technologies and instructional approaches without adequate support or training (Arvidsson et al., 2019; Jomoad et al., 2021; Munoz et al., 2025). Without sufficient time and resources, even well-designed innovations can become additional tasks rather than tools that enhance teaching and learning. In this study, generative AI is conceptualized primarily as a teacher-facing digital tool designed to support instructional planning and lesson development rather than as a student-facing classroom technology. This distinction positions AI as part of teachers' professional toolkit for creating, adapting, and reflecting on literacy-integrated science materials.

Recognizing these challenges, recent work has begun to explore how generative AI can serve as a practical means of reducing the time and effort required for instructional development. In professional learning contexts, AI tools such as ChatGPT have been used to generate lesson ideas, adapt instructional materials to curricular goals, and design classroom tasks—demonstrating their potential to streamline planning while maintaining alignment with disciplinary standards (Hu et al., 2024; Moundridou et al., 2024). At the same time, researchers emphasize that these tools require critical oversight to ensure the accuracy, credibility, and inclusiveness of AI-generated content (e.g., McCarthy & Yan, 2024). Emerging studies also highlight the growing importance of AI literacy, including

prompt engineering, as foundational skills for educators seeking to integrate these tools effectively (ElSayary et al., 2025; Sperling et al., 2024; Zhang & Zhang, 2024). Professional development programs that create opportunities for guided experimentation and reflection can help teachers balance efficiency with critical awareness, fostering the confidence and agency needed to use AI in pedagogically meaningful ways (Lan & Chen, 2024; Lu et al., 2024). Framing AI as a tool to support instructional design underscores teachers' dual role as creative users and critical evaluators of Gen-AI, ensuring that technology is used to enhance rather than replace their professional expertise (see Imundo et al., 2024).

1.4. Putting It All Together: The InSPECT Framework

Building on the instructional structures described above, the InSPECT framework complements lesson-planning approaches such as the BDA framework by focusing on the underlying cognitive and strategic processes that support multiple-document comprehension. Whereas BDA helps teachers organize and scaffold literacy instruction across a sequence of texts, InSPECT provides an instructional model that targets how students engage with and learn from those texts, specifically through sourcing and self-explanation strategies.

The InSPECT framework is grounded in theories of multiple-document comprehension (e.g., Perfetti et al., 1999; List & Alexander, 2019) and Self-Explanation Reading Training (McNamara, 2004, 2017), which itself is rooted in Construction-Integration theory (Kintsch, 1988). According to Construction-Integration theory, successful comprehension involves constructing connections within a text (the textbase) and integrating them with prior knowledge to form a coherent situation model. Self-Explanation Reading Training operationalizes these processes through generative strategies such as paraphrasing (restating text ideas in one's own words), bridging (linking ideas within and across texts), elaborating (connecting ideas to prior knowledge), and comprehension monitoring (evaluating one's understanding). These strategies promote inference generation and integration, which are core processes that distinguish shallow comprehension from deeper learning (McNamara, 2004, 2017; McNamara & McDaniel, 2004; O'Reilly & McNamara, 2007). By embedding these strategy routines within an instructional framework, InSPECT teaches students when and how to apply them during multiple-document reading and writing.

InSPECT bridges these perspectives to address a central limitation in existing multiple-document instructional approaches. While much existing work on multiple-document comprehension foregrounds sourcing, corroboration, and integration (e.g., Bråten et al., 2016; Braasch et al., 2013), these models often presuppose that students can already construct meaning from individual texts. In contrast, InSPECT begins with the recognition that students require both intratextual support to comprehend individual sources and explicit strategy guidance to build intertextual connections. By combining these two strands (i.e., sourcing from multiple-document comprehension research and self-explanation strategy instruction), InSPECT unites theoretical and instructional perspectives to support students in moving from understanding single texts to synthesizing ideas across multiple documents. These same theoretical foundations also guided the participatory design process, shaping how researchers and teachers collaboratively refined the framework and professional development materials through iterative cycles of reflection and classroom application.

InSPECT embeds core comprehension strategies drawn from Self-Explanation Reading Training (McNamara, 2004, 2017) within MD-specific practices, scaffolding students through the processes of evaluating sources, comparing perspectives, and constructing integrated representations. Rather than treating basic comprehension and multidocument reasoning as separate instructional goals, the framework treats them as mutually reinforcing. Figure 1 presents an overview of the InSPECT framework as introduced in professional development sessions.

InSPECT Framework

Integrating a self-directed technology-supported intervention in classrooms

In	Investigate	- Students make sense of their task and set goals. - What is required of them? What should they consider while reading? - Students should select and search for sources (if necessary).
S	Source Evaluate	- Before reading, students check the source type, verify the publisher, and verify the author. - While reading, students verify information accuracy across sources.
P	Paraphrase	- While reading, students write self-explanations by putting words from the text in their own words. - Students also monitor their comprehension while reading.
E	Explain	- Students also use the elaboration strategy to make connections and inferences about what the text is about. - Students continue monitoring their comprehension.
C	Connect Ideas Across Sources	- Students make connections using the bridging strategy. - Students verify information across sources. - Students continue monitoring their comprehension.
T	Transform your ideas to writing	- Students write summaries and reflections after reading sources. - After reading, students integrate multiple sources in a writing task. - Writing syntheses and helps students learn content and develop writing skills.

Figure 1. Slides from Professional Development Introducing the InSPECT Framework.

Consistent with other frameworks (e.g., MD-TRACE, IF-MT), InSPECT organizes tasks into phases that occur before reading, during reading, and after reading. This organization provides structure while also allowing for flexibility, as students may cycle back to revisit earlier strategies as needed. Figure 2 illustrates how these phases align with the Before–During–After (BDA) framework (W. E. Lewis & Strong, 2020).

Instruction before-during-after reading (B-D-A)

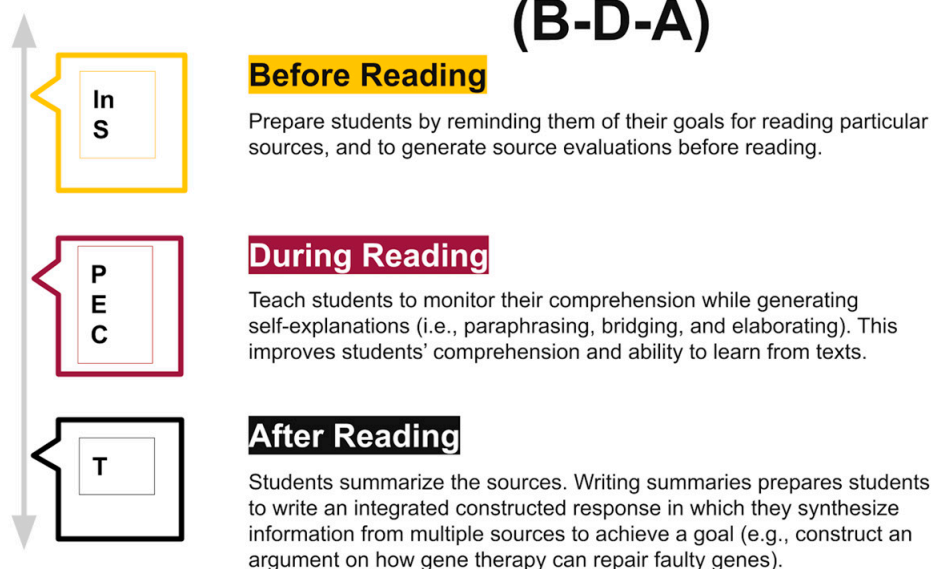


Figure 2. Slides from Professional Development Introducing the B-D-A Lesson Plan Framework.

1.4.1. Investigate

InSPECT begins with task representation and goal setting. Rather than treating reading as a passive intake of information, the Investigate phase encourages a goal-directed stance, aligning with research showing that stronger task models promote more effective selection, sourcing, and integration of information (Britt et al., 2018; List & Alexander, 2019). During

this phase, students ensure that they know the learning objectives associated with engaging in the task and the expected outcomes or deliverables.

1.4.2. Source Evaluate

After identifying potentially relevant texts, students engage in evaluating the quality and trustworthiness of each source. This includes considering the author's expertise, the publication venue, the date, possible biases, and the overall relevance of the source to their task and reading goals (e.g., [Brante & Strømsø, 2018](#)).

1.4.3. Paraphrase

Once reading begins, students engage in multiple strategies to support intratextual comprehension. Paraphrasing encourages students to restate ideas in their own words, ensuring accurate understanding and improving students' memory for the paraphrased information. Paraphrasing difficult words or phrases enhances students' comprehension of individual texts and prepares them to integrate information across sources ([McNamara, 2004, 2017](#)).

1.4.4. Explain

Explaining helps readers to connect ideas from sentence to sentence and invites them to access their prior knowledge to elaborate on the new information in the text. Elaboration helps readers understand how the new information connects to what they already know, and also to make connections with information from other texts in the text set ([McNamara, 2017](#)). The inferences generated through explanation support deeper comprehension by creating an elaborated and coherent intratext model and situating textual information within broader conceptual frameworks, rendering it easier for students to draw links across multiple documents.

1.4.5. Connect Ideas Across Sources

This phase emphasizes the development of an intertextual model, or a mental representation of how ideas relate across documents ([List & Alexander, 2019](#)). Students make connections and draw inferences both within individual texts and across multiple texts to strengthen this model (e.g., [Goldman, 2004](#); [McNamara et al., 2023](#)). They are encouraged to make explicit comparisons, question inconsistencies, and reason through possible explanations for agreement or disagreement among sources. This may involve evaluating the credibility of conflicting claims, reconciling apparent contradictions, or using one text to clarify or expand upon another. Intertextual integration extends beyond simply noting agreement or conflict; it also includes recognizing causal relationships, explanatory links, and thematic patterns that span across documents.

1.4.6. Transform Your Ideas into Writing

The final phase in InSPECT involves the transformation of understanding into written expression. Writing serves both as a product of learning and a process for organizing and consolidating information ([List & Alexander, 2019](#)). Students are guided to write brief summaries after reading each source to reinforce comprehension, and ultimately, to produce an integrated essay that synthesizes information across documents. [Figure 3](#) provides an example of how these strategies were represented in a text set-based lesson plan for classroom implementation.

Instructional Objective: Construct an argument for how gene editing can repair faulty genes.

Directions: Use the **InSPECT** Framework to write an argument for how gene editing can repair faulty genes. Read each of the sources below. Complete the reading guide to write **source evaluations**, **explain information**, **connect ideas across texts**, and **transform your ideas into writing**.

			
Audio-Visual	Background Info	Informational	Target Source
Link to source	Link to source	Link to source	Link to source
Reading Guide	Reading Guide	Reading Guide	Reading Guide

After reading each source, complete the **integrated essay**

Figure 3. Slides from Professional Development Introducing Example Lesson Plan.

1.5. Present Study

Implementing complex frameworks for reading, writing, and reasoning poses substantial challenges for teachers. Even when frameworks like InSPECT provide clear, research-based guidance, the realities of limited instructional time, competing curricular demands, and varied student readiness can make sustained implementation overwhelming (e.g., Blake & Traga Philippakos, 2025; Drew et al., 2017; Harris, 2024; Howell et al., 2021). Simply handing teachers a framework or set of materials is rarely sufficient; effective adoption requires collaboration, adaptation, and support structures that align with teachers' contexts and needs. As such, to help organize and implement these literacy practices more effectively, the BDA lesson-planning framework (W. E. Lewis & Strong, 2020) was introduced as a complementary structure to InSPECT to, provide a clear sequence for integrating strategies into teachers' unique contexts.

Moreover, to address these challenges, the current study examines how high school science teachers engaged as co-design partners in adapting the InSPECT framework for classroom use. While the larger project focuses broadly on developing and testing instructional approaches for multiple-document integrated reading and writing, this study centers on one phase of that work: a collaborative design process in which teachers and researchers worked together to adapt the framework, shape professional development activities, and iteratively refine instructional materials. As part of this process, teachers also experimented with generative AI tools to brainstorm lesson ideas, create adapted text sets, and design source evaluation tasks as a way to apply the InSPECT framework to their unique context. These activities allowed teachers to consider both the instructional potential of AI and the challenges of integrating such tools into science classrooms.

1. How do teachers collaboratively evaluate and shape PD activities and the InSPECT framework for supporting multiple-document integrated reading and writing?
2. What recommendations emerge from the co-design process for adapting PD materials and instructional practices to classroom contexts?
3. How do teachers perceive the opportunities and challenges of implementing the InSPECT framework, including the use of generative AI, in science classrooms?

2. Methods

2.1. Study Context

The current study is part of a larger, federally funded project supported by the Institute of Education Sciences (IES), which investigates effective approaches to supporting students in multiple-document integrated reading and writing tasks. Earlier phases of the project involved a series of lab-based studies that informed the development of the InSPECT framework. The current phase focuses on engaging teachers as co-designers through a series of focus groups and a pilot implementation of the framework. These sessions marked a shift from researcher-led development to participatory design, positioning teachers as active contributors in refining and extending InSPECT to meet the needs of their students and instructional contexts.

2.2. Participants

Sixteen high school biology teachers from Georgia ($n = 4$) and Ohio ($n = 12$) participated in the study. Thirteen participants were women and three were men. Most participants ($n = 11$) reported having taught all high school grades (i.e., 9–12th) in the past three years. Teachers reported experience across multiple science disciplines, most commonly biology ($n = 9$), environmental science ($n = 8$), physical science ($n = 7$), anatomy ($n = 7$), and honors or advanced biology ($n = 4$). Table 1 provides participant demographics with pseudonyms used in the findings.

Table 1. Participant Background.

Pseudonym	State	Grades Taught (Past 3 Years)	Gender	Science Areas Taught (Past 3 Years)
Alicia	GA	9–12	Female	Biology, Environmental Science
Bianca	OH	9–12	Female	Biology, Physical Science, Environmental Science
Claire	OH	10–12	Female	Advanced Biology, Environmental Science, Chemistry, Anatomy (study assist)
Carly	OH	9–12	Female	Biology, STEM, Anatomy, Environmental Science, Forensic Science, Advanced Biology
Dana	OH	9–12	Female	Biology, IB Biology, AP Biology, Anatomy & Physiology
Emma	OH	9–12	Female	Multiple science courses (alternative high school context)
Erin	OH	9–12	Female	Biology, Physical Science, Environmental Science
Isaac	OH	9–12	Male	Physical Science, Chemistry, Biomedical Science, Materials Science
Jordan	GA	9–12	Male	Biology, Honors/Gifted Biology, AP Biology
Jenna	OH	9–12	Female	Physical Science, Earth Science
Kayla	GA	11–12	Female	Human Anatomy & Physiology, Epidemiology, Forensics
Molly	OH	9–12	Female	Physical Science, Environmental Science, Anatomy & Physiology, General Science (Project-Based Learning)
Priya	OH	10–12	Female	Biology, Environmental Science
Riley	OH	9–12	Female	Biology, Environmental Science, Tutor in Physical Science
Ryan	OH	9–12	Male	Physics, Physical Science, Biology
Yasmin	GA	10–11	Female	Physics, Anatomy & Physiology

2.3. Research Design

This study employed a participatory, mixed-methods design (Creswell & Plano Clark, 2017) to engage high school science teachers as co-designers in adapting and extending two instructional frameworks (i.e., the InSPECT framework for multiple-document comprehension and the BDA lesson planning structure). The design goals were to (a) refine the frameworks for use in secondary science classrooms, (b) identify professional development features that could introduce the frameworks to new teachers and support their adaptation to diverse contexts and communities, (c) co-develop and revise PD activities, (d) pilot and share instructional materials, and (e) explore the role of generative AI tools in lesson planning and material design.

A participatory design framework guided the study. Participatory design emphasizes the direct involvement of users in shaping the development and evaluation of tools and systems (Muller & Kuhn, 1993). As Wacnik et al. (2025) note, participatory design encompasses varied user-centered practices that differ in timing, techniques, and stakeholder involvement. In this study, teachers contributed at early and mid-stage leverage points through iterative focus group engagement and a prototyping pilot, which enabled them to refine PD activities and instructional materials while adapting the InSPECT framework. Prior work demonstrates that participatory design has been used with teachers to develop both instructional practices and educational technologies (Cumbo & Selwyn, 2022; Tuhkala, 2021). Co-design, as one form of participatory design, emphasizes teachers' sustained collaboration with researchers to ensure that innovations fit classroom contexts (Roschelle & Penuel, 2006). Recent work has further shown how co-design can be used with teachers to integrate AI into instructional materials and explore new classroom applications (Kosmas et al., 2025). Consistent with this literature, our study engaged teachers in refining evidence-based literacy practices and experimenting with generative AI tools to support lesson design.

2.4. Study Procedures

2.4.1. Focus Group Sessions

The study consisted of five semi-structured focus group meetings designed to engage teachers in the co-design of a professional development program. Each session lasted approximately 90 min and was held virtually via Zoom. All sessions were recorded, and participants provided informed consent prior to participation. Sessions were intentionally structured so that teachers first experienced sample PD activities (e.g., introductions to frameworks, practice with instructional strategies, or collaborative lesson design) and then reflected on their usefulness, revisions, and challenges. This design allowed teachers to participate both as learners and as co-designers, providing iterative feedback on what to keep, revise, or remove from the PD model. Across all sessions, breakout discussions were followed by whole-group synthesis, and teachers were assigned preparatory tasks between sessions (e.g., reviewing texts, standards, or lesson materials).

Session 1. Facilitators introduced the project, reviewed study procedures, and established group norms. Teachers engaged in a brainstorm activity to identify components of PD they had previously found beneficial and features they viewed as ineffective or counterproductive. In breakout groups, they generated lists of "keep" and "avoid" features, then shared back to the full group. This session concluded with participants ranking these features in a survey.

Session 2. The session began with a recap of themes from Session 1 and a preview of upcoming PD content. Teachers participated in activities designed to build a shared foundation for disciplinary literacy instruction: defining reading comprehension and multiple-document comprehension, and reviewing evidence-based strategies for supporting students (paraphrasing, bridging, elaboration, and source evaluation). Teachers then

previewed sample text sets and accompanying videos that modeled these strategies, and were introduced to the InSPECT framework. Activities also included discussion of how such strategies could be integrated into science classrooms. A generative AI component invited teachers to brainstorm classroom uses of ChatGPT (OpenAI, 2025, version 3.5) and engage in an introductory prompt-engineering activity. Teachers completed surveys rating the usefulness of these PD features.

Session 3. Following a recap of Session 2, teachers continued reviewing intervention materials, including self-explanation and source evaluation strategies embedded within reading guides. A key activity centered on assessment practices—how self-explanations might be scored, and how generative AI could assist with or complicate that process. Teachers then engaged in a structured prompt-engineering activity in which they used ChatGPT (version 3.5) or Gemini (formerly Google Bard; Google DeepMind, 2025) to create a set of example sources for students to evaluate. Teachers were tasked with (a) generating sources that varied in credibility, (b) adapting sources to their own curricular content, and (c) developing lesson procedures for students to practice source evaluation. Breakout groups compared outputs, discussed challenges in using Gen-AI for lesson planning, and reflected on questions and concerns for classroom use.

Session 4. After reviewing prior themes, the session turned to a deep dive into the InSPECT and BDA frameworks for lesson planning. Teachers reviewed a sample biology lesson plan and then worked in breakout groups to consider benefits, challenges, and classroom applications of each framework. Discussion prompts included: What are the benefits of teaching the InSPECT framework? What challenges might arise in implementing it? How does the BDA format compare to teachers' current lesson design approaches? What challenges might teachers encounter with BDA? The purpose of these discussions were to examine the extent to which the frameworks could support disciplinary literacy, and how they might need to be adapted for science-specific contexts.

Session 5. The final session began with a recap of Session 4 themes and transitioned into collaborative lesson building. Teachers had structured work time to design their own lessons using InSPECT and BDA templates, adapting materials to their disciplinary content. They then participated in a PD retrospective activity, sharing feedback on what to keep, revise, remove, or add to the PD program. This concluding activity surfaced targeted recommendations for refining the PD model.

2.4.2. Pilot Implementation

Following the focus group series, one participating teacher piloted the co-designed instructional materials in her high school biology classroom. She implemented a sequence of five lessons in which students were introduced to the InSPECT framework and applied it within a biology unit. Over the course of the unit, students read four source texts, took structured notes on each, and engaged in comprehension activities guided by the framework. The sequence concluded with students writing an integrated essay synthesizing information across the four sources.

2.5. Data Sources

Data were collected from multiple sources to capture teacher perspectives during the co-design process and outcomes of the pilot implementation.

2.5.1. Quantitative Survey Data

At the start of Session 1, participants completed a survey to capture baseline familiarity with ChatGPT. The remainder of the quantitative data was teacher feedback on the session activities. In each focus group session, teachers completed surveys in which they ranked or rated the usefulness of PD features and activities. Usefulness ratings were collected

on a 5-point Likert-type scale (1 = not at all useful to 5 = extremely useful), and feature rankings were captured using a forced-order ranking format. After Session 1, participants ranked positive and negative features of PD that were discussed during the session. In Sessions 2 and 3, participants responded to a survey to rate the usefulness of specific PD activities (introducing and explaining the InSPECT and BDA frameworks, review of self-explanation and source evaluation instructional materials, lesson-planning templates, and a brainstorm activity using the digital collaborative white board, Miro Board). In Sessions 4 and 5, participants participated in PD activities and then rated their usefulness, including framework discussions, dedicated work time to develop their own lesson plans, and review of example lesson plans. Participants also responded to survey items regarding Gen-AI activities (e.g., reviewing example ChatGPT outputs for lesson materials, brainstorming classroom uses, and prompt-engineering practice).

2.5.2. Transcript Data

Focus group and interview transcripts provided qualitative data on teacher perspectives. Transcripts from the five focus group sessions captured teachers' perceptions of the professional development, design recommendations for improving the PD and InSPECT frameworks, and considerations for classroom implementation. The interview with the pilot teacher captured her reflections on the usefulness of the PD, challenges encountered during implementation, observed impacts on students, and recommendations for refining the program and framework.

2.5.3. Student Work Samples

We also had consent to analyze 8 students' work from the pilot study in Dana's 10th-grade biology class. For the pilot implementation, Dana selected the sample biology lessons on CRISPR gene editing rather than developing new instructional materials. She explained that implementation occurred near the end of the academic year, following state testing, which allowed greater flexibility in curricular choices. She selected the CRISPR lessons because she believed the topic would be both relevant and engaging for her students. The work included structured notes on four source texts and a culminating integrated essay to provide evidence of how students engaged with the InSPECT framework and synthesized information across multiple sources.

2.6. Data Analytic Plan

We employed a convergent concurrent mixed-methods design (Creswell & Plano Clark, 2017) in which qualitative and quantitative data were collected concurrently, analyzed independently, and integrated for interpretation. This design prioritized qualitative analysis (QUAL + quan), centering teachers' voices from focus group transcripts, the pilot teacher interview, and student work samples, while drawing on survey data to triangulate themes (Fielding, 2012; Toledo & Shannon-Baker, 2023). Integration occurred during interpretation and reporting, where qualitative themes were aligned with corresponding quantitative survey items (e.g., teacher ratings of framework clarity, relevance, or feasibility) and presented in joint displays to highlight areas of convergence and divergence between data sources. Instances where qualitative narratives and survey trends diverged were examined to contextualize differences and refine interpretation.

Qualitative analysis followed a general inductive approach (Thomas, 2006) with structured coding and synthesis strategies (Saldaña, 2014). The unit of analysis was individual speaker turns or meaning units that reflected a complete idea. Consistent with Thomas (2006) procedure, analysis proceeded through iterative stages of data preparation, close reading, category creation, overlapping coding, and refinement. Transcripts, interview notes, and student artifacts were first formatted and reviewed multiple times to gain famil-

ilarity with the content and identify recurring ideas. Initial categories were developed both a priori (derived from evaluation aims such as relevance, feasibility, and usefulness of PD) and inductively from participants' language (e.g., templates ease planning, paraphrasing as scaffold, source-evaluation challenges, AI as planning aid, and concerns about misuse). Codes were first descriptive (e.g., "clarity of framework," "classroom feasibility") and later synthesized into broader interpretive themes (e.g., "practicality and alignment," "collaborative design value"). Theme definitions were refined through iterative review to ensure each was internally consistent and distinct. Two researchers independently reviewed all qualitative sources, generated initial codes and representative quotations, and then met to compare codebooks, resolve discrepancies, and consolidate categories into final themes.

To support efficiency, we used a qualitative analysis tool built with GPT-4o (Potter et al., 2025) to assist with identifying potential themes, condensing meaning units, and sourcing illustrative quotations for the themes using the transcript data. Final themes were defined by the researchers, ensuring interpretive accuracy and alignment with study questions. The LLM was not relied upon to independently code data or generate themes, but to complement researcher-led analysis and interpretation. Human-AI workflows increase efficiency, while preserving rigor and researcher control (Hayes, 2025). To ensure trustworthiness of the qualitative analysis, discrepancies were resolved through consensus, and findings were triangulated across focus groups, interviews, surveys, and student work.

Student artifacts from the pilot classroom were examined to document engagement with the InSPECT framework, including completeness of notes and evidence of synthesis in essays.

Quantitative analysis involved descriptive statistics of survey data to assess trends in teachers' ratings of PD activities and frameworks. After separate analyses, findings were integrated using a side-by-side comparison approach (Creswell & Plano Clark, 2017) to identify areas of convergence and divergence between qualitative themes and survey results.

3. Findings

The analysis of survey data, focus group transcripts, the pilot teacher interview, and student work artifacts generated five themes: (1) Practical, Relevant, and Feasible PD, (2) Usefulness of InSPECT and BDA Frameworks, (3) Supporting Deeper Strategies Beyond Paraphrasing, (4) Integrating Gen-AI into PD and Practice, and (5) Classroom Implementation Challenges. Below, we present each theme with supporting quantitative and qualitative evidence, followed by interpretive commentary.

3.1. Theme 1: Teachers Evaluated the Professional Development in Terms of How Practical, Relevant, and Feasible It Was for Their Classroom Practice

Teachers emphasized that professional development was most effective when it was concise, directly connected to their subject matter, and respectful of their limited time. In the opening sessions of the PD, teachers were asked to reflect on their prior experiences with professional development and to identify components that made PD valuable as well as those that undermined its usefulness. Their responses consistently emphasized the need for PD that was relevant, hands-on, and time-efficient. As one teacher wrote in the focus group reflection activity, "PD that works well is PD that is relevant and is useful to us teachers in the classroom".

Teachers also underscored how the lesson planning templates and work time aligned with their call for PD that was hands-on and immediately usable. During a collaborative focus group session, Molly explained, "The templates definitely help ease the planning load". In the same discussion, several teachers highlighted the value of having both models and opportunities to practice. Dana commented on the usefulness of dedicated work time, while Isaac emphasized that "Giving teachers a chance to view or participate in new

teaching approaches from a student perspective is incredibly valuable". Building on this point, Riley added in the same focus group session, "I appreciated that you had an example [lesson plan and materials] and then gave the opportunity for us to try [and create our own lessons]". Similarly, Priya stressed in the focus group that the PD was meaningful precisely because it centered on science teachers' needs, noting, "I'm tickled that writing is the end product of this. I want my students' writing to improve for science. I also appreciate that this was science-based PD so everyone understands each other".

At the same time, teachers noted that there was still room for improvement in making the PD more directly relevant to their classroom contexts. While they valued the structured frameworks and opportunities for practice, several emphasized that examples and materials should be more explicitly tied to disciplinary content. Emma pointed out during a focus group discussion that she wanted "examples that are current issues in science," underscoring the importance of disciplinary relevance. In the same discussion, Isaac added that he would like "more standards-based science examples—not just generic ones, but examples from Physical Science, Chemistry, Conservation of Energy, [and] Gas Laws." Although teachers indicated the PD was already relevant and practical, teachers wanted it to be anchored even more firmly in science-specific content relevant to their contexts.

These qualitative reflections align closely with the survey responses. In Session 1, teachers first identified the features of effective and ineffective PD during the focus group discussion and then ranked these same features in a follow-up survey. Consistent with their qualitative reflections, teachers rated relevance, engaging presentation, and opportunities to practice as the most important features of effective PD, while identifying irrelevant content, lack of practice, and disengaging presentations as the most important shortcomings to avoid (see Table 2).

Table 2. Survey Rankings of Professional Development Features and Teacher Reflections.

PD Feature	Mean Rank ¹	Representative Teacher Quotation
Positive Features		
Relevant to my teaching context	1.53	"PD that was beneficial to my content area and gave me things to take back to the class."
Engaging presentation	2.87	"Hands on PD that kept me engaged and doing activities."
Opportunities to practice new skills	3.13	"We had time to practice during the PD."
Negative Features		
Irrelevant content	2.13	"Having to go to a PD that had nothing to do with my content area."
Lack of practice	4.33	"No interacting. No 'try it' component. Just... 'here you go.'"
Not engaging	4.4	"Having somebody just read off the slides to us."

¹ Positive features were ranked from 1 (most important) to 8 (least important). Negative features were ranked from 1 (most important to avoid) to 9 (least important to avoid). Lower mean rank indicates greater importance. Representative quotations were drawn from the written reflection exercise completed immediately following the focus group discussion of professional development features.

3.2. Theme 2: Teachers Described How the InSPECT and BDA Frameworks Supported Their Instruction by Providing Clear, Structured Approaches to Disciplinary Literacy

Teachers explained that the InSPECT and BDA frameworks offered clear and structured guidance for planning instruction, supporting students' reading and writing development, and differentiating tasks for varied learners. Dana, the teacher who implemented the pilot lessons, reflected during the focus group session that the frameworks would not be challenging to use because they "laid out the steps I needed to take in order to help my students." She elaborated,

As a science teacher, I maybe shouldn't admit this to this group, but sometimes with students who struggle with reading, I struggle in knowing the steps to take to help them. This [framework] made it easy: This is what I do, this is what they do, in these steps.

After piloting the InSPECT lessons, Dana also explained that the framework “kept them moving in a chronological type pattern. . . to get to the end result” and that she “found it really easy. The kids caught on really quick, and how to use it.” Her comments reinforced the frameworks’ step-by-step organization supported both instructional planning and student comprehension.

Other teachers in the focus groups echoed this point. Priya remarked that the frameworks were “easy [to implement] because of the steps,” while also acknowledging that “I too struggle with struggling readers. Sometimes it is very difficult to know how to help them.” Another teacher reflected that “My students struggle so much with prior knowledge and making connections to their own experiences because they are so lacking in both.” These reflections underscored how the frameworks provided a clear process that teachers could adapt to students with diverse literacy needs.

Teachers also expressed that the lesson-planning templates accompanying the frameworks helped them see immediate classroom applicability. Molly explained in the focus group sessions that, “The templates definitely help ease the planning load,” while Jenna noted that she appreciated the examples provided. At the same time, participants emphasized the importance of balancing structure with flexibility in professional learning. Isaac noted in the focus group that the BDA lesson planning “felt redundant,” suggesting to “Keep it on templates, but don’t focus much on it in PD.”

Survey data reinforced these qualitative reflections. After Sessions 4 and 5, teachers rated framework explanations, discussions, and example reviews as highly useful, while brainstorming activities using the Miro board were rated less useful. Open-ended survey responses emphasized the value of lesson planning templates and dedicated work time, highlighting that these supports made the frameworks both clear and immediately actionable for classroom practice (Tables 2 and 3).

Table 3. Survey Ratings of PD Activities and Representative Reflections.

PD Activity	M (SD)	Representative Teacher Reflection (Open-Ended)
Review InSPECT lesson framework	3.88 (0.34)	“It really helps to see how this might be implemented with students. You obviously have a very well-developed system.”
Review of BDA lesson planning framework	3.63 (0.62)	“The BDA gives students a focus during reading. . . sets them up with background, a purpose, smaller chunks, and a follow-up. It’s an ideal way to take in information.”
Review of lesson planning templates	3.69 (0.60)	“The way the template was broken down was really beneficial—it made it easy to see where everything belonged.”
Work time to develop BDA lesson objective	3.69 (0.60)	“Getting to work with the templates to start putting lessons together while asking questions and sharing thoughts was most helpful.”
Work time to develop text sets for BDA lessons	3.69 (0.60)	“I really enjoyed creating the lesson plan and materials—I’ll be using this in my class in a few weeks.”
Work time to develop completed BDA lesson plan	3.63 (0.62)	“Applying it to my own course let me think much more deeply about it.”

Note. Ratings were provided on a scale from 1 (not useful at all) to 4 (very useful). Means and standard deviations are shown in parentheses. Representative reflections are drawn from open-ended responses and exemplify the prevailing sentiment for each item.

3.3. Theme 3: Teachers Noted That Students Primarily Used Paraphrasing, While Other Strategies from the InSPECT Framework Required Additional Instruction and Practice

Teachers emphasized that paraphrasing was the most accessible strategy for students and served as a scaffold for their writing. Dana, the teacher who implemented the pilot lessons, reflected in her post-implementation interview that “paraphrasing was really helpful for them” and directly impacted their writing performance: “Their essays were fantastic. . . once they had all that paraphrasing, they were able to pull that together and actually write better essays.” She went on to explain that while her students sometimes had writing opportunities in biology, these were usually short and less sustained: “It’s more short writing. . . not to that extent.” Dana described being genuinely impressed by the quality of several final essays, noting that “a couple of students who, after going through all of them, and then they submitted their essay. . . they were fantastic. Their essays were fantastic. And I said, wow, like, just, wow! I don’t know what else to tell you.” Her students also valued the opportunity to engage in more extended writing: “They said, ‘I liked this opportunity to express myself in writing because they got to take multiple things, put them together and then write about it,’ which they said was their strength. We do writing in biology, but not to that extent. . . it’s more short writing.” These reflections indicate that while paraphrasing initially helped students process information from individual sources, it also provided a planning tool to help integrate ideas across texts and improve their writing. Lessons from the InSPECT framework, therefore, provided a structure that both supported comprehension and gave students an opportunity to engage in more extended writing than was typical in their science coursework.

Early reading guides from one of Dana’s students, whom she later described as producing a “fantastic” final essay, showed an initial reliance on paraphrasing short segments of text (see Supplement S1 ([Mayo Clinic, 2021](#))). For example, the student wrote, “CRISPR/Cas9 is made up of two main components: The Cas9 protein (preps DNA) and a guide RNA (recognizes sequences of DNA that need to be edited).” By the fourth source, however, the same student demonstrated elaboration (“I’ve heard of this disease before and how it affects people’s lives heavily. . .”) and bridging (“The difference, however, is that the parents typically don’t show symptoms of this disease”; see Supplement S2 ([MedlinePlus, n.d.](#))). The student’s final essay further illustrated this progression, combining accurate paraphrases of scientific information with elaboration and synthesis across sources (see Supplement S3). Using the four source texts provided in the InSPECT lesson set, the student synthesized information about CRISPR/Cas9 to explain how it “correct[s] gene sequences in our DNA to treat a disease.” and linked these explanations to broader issues of medical research, noting that “not knowing what awaits shouldn’t be a reason to not advance.” These artifacts show how repeated practice with paraphrasing provided a foundation for comprehension and later supported the integration and evaluation of ideas in writing.

While paraphrasing supported student writing, teachers agreed that other InSPECT strategies, such as source evaluation, bridging, and elaboration, were more difficult for students without additional modeling and practice. In her interview, Dana reflected, “Some of the students had a hard time verifying the publisher. They didn’t understand what that meant. They would just leave it blank or write ‘good.’ I had to say, ‘Guys, you have to look at the publisher. How do you know if it’s good?’ That was probably the biggest challenge.” She also noted that while some students attempted to make bridging connections between texts, most needed more time and explicit scaffolding: “Some students did a really good job making connections. . . but I think if we had had more time, we could have discussed elaborations more, just to get them out of there.”

Teachers in the focus groups echoed these insights. Kayla remarked in the focus group that her biggest instructional challenge was that “kids don’t know how to paraphrase,

bridge, or elaborate. I teach juniors and seniors and still have to show them how to paraphrase and take notes. They just copy PowerPoint slides or take pictures.” Similarly, Yasmin then explained her concern about synthesis writing: “A student might just take a point from one source. I’ll be like, ‘That’s not synthesis, you have to synthesize across sources.’ That’s my challenge. If a source is extensive, or there are multiple sources, they tend to just pull from one.” Across teachers’ reflections, paraphrasing emerged as an essential first step that enabled students to comprehend and engage with complex texts, yet teachers consistently noted that strategies such as bridging, elaboration, and source evaluation required continued modeling and guided practice. These findings suggest that subsequent iterations of the framework should include stronger scaffolds and examples for teaching higher-level comprehension strategies.

3.4. Theme 4: Teachers Reflected on the Potential of Generative AI Tools to Support Lesson Planning and Classroom Instruction, While Also Raising Concerns About Feasibility and Misuse

Teachers were intrigued by the potential of generative AI tools, particularly LLMs such as ChatGPT, to make lesson planning and material preparation more efficient. Jenna said in the focus group that it could be useful when crunched for time: “I could definitely see using this when I need something real quick, or maybe when I have to have a sub.” Other teachers echoed this point, noting that AI could generate draft materials that they could then adapt for classroom use. Beyond efficiency, participants also imagined instructional applications. In the focus group, Dana suggested generating versions of texts at different Lexile levels so students could compare them, while Emma proposed having students fact-check AI outputs for “fact vs. opinion” exercises. Isaac extended this idea, suggesting ChatGPT could provide a mix of credible and non-credible science links for source evaluation, and also envisioned using it to generate prompts or scaffold student responses (e.g., “Give me an intro sentence on X topic”). These reflections highlight how teachers viewed LLMs as both a planning tool and a scaffold for student practice.

Survey data reinforced these qualitative insights. Teachers rated ChatGPT activities in the PD, including seeing example outputs, brainstorming applications, and practicing prompt engineering, as highly useful overall (see Table 4). Notably, most participants began with little to no prior familiarity with LLMs, yet still found the activities valuable.

Table 4. Survey Ratings of ChatGPT PD Activities and Teacher Reflections.

PD Activity	M (SD)	Representative Reflection
Seeing example of ChatGPT text	3.73 (0.46)	“Actually seeing how the Chat worked and trying it myself. . . I thought it was pretty cool.”
ChatGPT brainstorming activity	3.33 (0.72)	“I loved the brainstorm of how ChatGPT can be used but would have liked more time to look at others’ answers.”
Prompt engineering activity using ChatGPT	3.63 (0.72)	“The prompt activity using ChatGPT. . . I think it was useful and actionable.”

Note. Ratings were provided on a scale from 1 (not useful at all) to 4 (very useful). Teachers’ average familiarity with ChatGPT before the PD was low ($M = 2.00$, $SD = 1.03$ on a 4-point scale). Representative reflections are drawn from open-ended survey responses.

At the same time, teachers raised significant concerns about misuse. During the focus group, Jenna worried that “a kid [might] say, ‘Oh yeah, I wrote this,’ and it’s totally something they got from [ChatGPT]. That’s my concern—the plagiarism.” Bianca echoed this point, observing that students already struggle with citation and might simply copy and paste outputs. Teachers therefore stressed the importance of clear norms for ethical use. Emma framed this as a professional development need, suggesting that teachers should

learn how to “set expectations with students on AI use, especially around ethics, cheating, and plagiarism.” Molly added, “Trying to keep them from using it is a lost cause. . . I think teaching them to use it right, teaching them the appropriate time to use it.” Finally, Isaac noted that while AI could handle many routine tasks, it could never replace the core of teaching: “It can do a lot of the mundane and generic parts of our jobs, but it won’t be able to replace the truly valuable things we do—building relationships, challenging students to think deeper, provoking critical thinking.”

3.5. Theme 5: Teachers Identified Persistent Challenges in Implementing InSPECT, Including Limited Time, Grading Constraints, and Uneven Student Readiness

While teachers saw value in the InSPECT framework, they also raised concerns about barriers to implementing it consistently in their classrooms. Dana noted in her interview following the pilot implementation that the lessons themselves “didn’t take that long” and could be integrated into a unit without difficulty. However, other teachers anticipated that instructional time would be the most significant challenge, particularly when trying to fit multi-step activities into 45–50 min class periods. Molly explained in the focus group that “This will not be a short activity—this is a week, probably, from start to finish. But we’re always looking for different types of assessments or things to throw in. . . I like that it could break things up”. Molly and Ryan suggested implementing InSPECT once a quarter. Similarly, Claire shared in the focus group that “I think it’s wonderful. . . but I was thinking about whether it should be a part of every unit, or whether it should be implemented more sparingly, to keep it fresh”.

Teachers also pointed to workload as a barrier, especially at the start of implementation. Molly explained, “As with new initiatives, this looks like a lot of front-loaded work, but the templates will make it usable.” She reiterated that, “there’ll be some front-loaded lessons on the process, which are easy to add to start-of-year work.” These comments suggest that while teachers expected an initial learning curve, they also believed the templates provided would reduce the burden and make InSPECT feasible over time. Teachers also identified grading and assessment as a persistent concern. Claire asked directly during the focus group, “I’m interested in how you assess this, or how it relates to grades”. Participants worried about the burden of evaluating multiple student products, particularly with large class sizes. While some suggested adapting existing reading and writing rubrics, others highlighted that holistic scoring or clustering responses might be more feasible than grading each artifact individually.

Another barrier was perceived variance in students’ reading and writing abilities which may require additional scaffolding to implement the InSPECT framework. Indeed, Claire shared that “Some of my students, especially ninth graders, would get overwhelmed if this was introduced all at once. They’d need more modeling and smaller steps.” Molly also noted that even though the InSPECT framework is designed to build students’ prior knowledge on a topic before reading an advanced text (i.e., the “target source”), she said that nevertheless these “sources are tough because of the wide variety of reading levels in our students”. Molly went on to say that “the students willing to do it would likely be high-achieving ones.”

Concerns about uneven student readiness also extended to equity and participation, particularly regarding challenges related to technology access. Jordan pointed out during the focus group discussion that “[T]hese activities seem very geared toward an online classroom. . . but our district is not going to give students laptops back this year. So they want us doing more paper-and-pencil type things. How could a teacher incorporate something like this with less technology access?”. Claire responded that teachers might adapt by using print resources and classroom stations: “If you have trade magazines aligned with your curriculum, those could be used. If students had stations where they

could go—we often print articles and set up stations”. Bianca added that “Some [students] don’t have computer access outside school,” highlighting structural barriers that could further limit participation.

4. Discussion

This study examined how high school science teachers, engaged as co-design partners, evaluated and refined a professional development program centered on frameworks to support multiple-document integrated reading and writing. Using a participatory mixed-methods design, the analysis addressed three research questions focused on (RQ1) how teachers evaluated the quality and design of the PD and its frameworks, (RQ2) what recommendations emerged from the co-design process for adapting PD materials and practices, and (RQ3) how teachers perceived the opportunities and challenges of implementing the InSPECT framework and using generative AI in science classrooms.

Five findings emerged. Findings related to RQ1, which examined how teachers evaluated the quality and design of the PD, showed that teachers valued PD that was practical, relevant to science content, and feasible within their time constraints, particularly when it offered templates, models, and structured work time. Addressing RQ2, which focused on recommendations from the co-design process, teachers viewed InSPECT and BDA as clear, stepwise scaffolds that made disciplinary literacy instruction more actionable, while requesting more science-specific exemplars. Also related to RQ2, teachers noted that students most readily used paraphrasing, whereas other strategies (i.e., source evaluation, bridging, and elaboration) required more explicit modeling and practice, which should guide subsequent PD revisions. Findings tied to RQ3, which explored teachers’ perceptions of opportunities and challenges in implementation, indicated that teachers perceived Gen-AI as a promising tool for planning and for designing literacy tasks, while emphasizing the need to address ethics, plagiarism, and classroom norms. Finally, RQ3 findings also highlighted anticipated barriers to implementation, including limited instructional time, grading load, variable student readiness, technology access, and equity concerns.

4.1. Interpretation of Findings

Teachers emphasized the importance of professional development that was practical, relevant, and immediately usable, highlighting conditions that support the integration of literacy practices into science instruction. They identified concise sessions, alignment with science content, and structured opportunities for practice as essential, and called for more discipline-specific exemplars tied directly to standards and classroom topics. These reflections echo prior work showing that science teachers often feel underprepared for literacy instruction but benefit from PD that builds their capacity to integrate literacy routines into science instruction (Blake & Traga Philippakos, 2025; Lupo et al., 2023; Thomas & Drew, 2021). Consistent with this emphasis on feasibility, Stevens et al. (2022) found that even modest, flexible professional development structures improved student outcomes when integrated across content instruction.

The InSPECT and BDA frameworks illustrated how such principles could be operationalized. Teachers described them as clear and stepwise, with templates that reduced planning load and made literacy instruction more manageable. At the same time, they noted areas for refinement, such as offering more science-specific exemplars and clearer alignment with teachers’ daily routines. Evidence from the classroom pilot further confirmed their feasibility, as the participating teacher reported straightforward implementation and observed positive effects on student writing. These findings suggest that teachers value PD that balances generalizable frameworks with concrete, discipline-based demonstrations of practice,

reflecting some of the core design principles of effective PD (e.g., [Darling-Hammond et al., 2017](#); [Thomas & Drew, 2021](#)).

Teachers reported that paraphrasing was the most accessible entry point for students, providing a foundation for more complex strategies in the InSPECT framework. Although paraphrasing is often associated with surface-level understanding, it is a critical step in enabling students to engage in bridging, elaboration, and intertextual connections ([McNamara et al., 2009, 2023](#)). Proficiency in these higher-order strategies requires additional modeling, guided practice, and sustained opportunities to apply them ([Barzilai et al., 2018](#); [McNamara, 2017](#)). Yet teachers also pointed to barriers that make such sustained work difficult, including limited instructional time, grading demands, uneven student readiness, and technology access. Even though the BDA routines are designed to be flexible, they nonetheless require consistent implementation over time to support growth ([W. E. Lewis & Strong, 2020](#); [Lupo et al., 2023](#)). Teachers anticipated that adoption of the InSPECT framework would therefore need to be gradual and supported with streamlined assessments and differentiated scaffolds. Research further suggests that sustaining such efforts requires shared professional learning among teachers and administrators to build a collective vision for literacy across disciplines ([Lupo et al., 2023](#); [Thomas & Drew, 2021](#)). Thus, implementing frameworks like InSPECT depends not only on individual teacher effort but also on leadership support and coordinated engagement across teacher teams over time.

In light of these challenges, teachers considered the potential role of generative AI in addressing some of the practical barriers to implementing literacy instruction. Their interest in using AI to rapidly generate or adapt materials, paired with concerns about plagiarism and credibility, mirrors emerging evidence: AI can accelerate lesson design but still requires teacher oversight, AI literacy, and critical evaluation of outputs ([ElSayary et al., 2025](#); [Hu et al., 2024](#); [Moundridou et al., 2024](#)). Positioning AI as both a design aid for teachers (e.g., drafting text sets, differentiating materials) and as a source of outputs for students to evaluate and fact-check reflects the same PD design principles that made InSPECT and BDA useful: providing concrete tools while also building professional judgment. Embedding AI-focused activities such as prompt engineering and output analysis into PD can therefore extend teachers' capacity to integrate literacy instruction efficiently while fostering students' critical literacy, provided that explicit classroom norms for ethical use are taught ([Lan & Chen, 2024](#); [Lu et al., 2024](#)).

4.2. Implications and Revisions to Professional Development

This project was grounded in a participatory design approach, with the purpose of refining the professional development model so that it could be implemented more effectively in secondary science classrooms. Teachers acted as co-designers who shaped both the instructional frameworks and the PD activities, ensuring that revisions addressed their practical needs and contexts. Teachers consistently requested more discipline-specific exemplars aligned with science standards. While it is not feasible to produce exemplars for every domain, we developed additional content examples and built in collaborative work time so teachers could co-develop lessons with colleagues in similar content areas. This approach provides both ready-to-use materials and opportunities for teachers to localize practices.

Revisions also focused on integrating the InSPECT and BDA frameworks into existing routines while addressing structural barriers such as time, workload, and uneven readiness. The PD now includes opportunities for teachers to share the literacy strategies they already use, identify ways to align them with InSPECT, and adapt routines to daily schedules and curricular pacing. We emphasized collaboration among teacher teams from the same

schools so they could coordinate pacing, grading, and scaffolding in ways that fit local contexts. These changes ensure that new practices are embedded within existing routines rather than added on top of them.

A third revision emphasized helping teachers move beyond paraphrasing toward more advanced strategies such as elaboration, bridging, and source evaluation. Teachers identified paraphrasing as accessible but stressed that students needed additional modeling and practice with higher-order strategies. In response, the PD devoted more time to these strategies and provided supplemental resources and instructional videos from prior research. These additions give teachers concrete materials to support instruction and scaffold student development across the school year.

Finally, teachers' reflections on generative AI underscored both opportunities and challenges. They saw potential for AI to reduce planning time and generate adaptable materials, while also raising concerns about plagiarism, accuracy, and ethics. Because the focus groups took place in summer 2023, teachers may now have more classroom experience with AI; nevertheless, this remains a critical area for PD. Moving forward, we will embed modules on AI literacy that prepare teachers to evaluate outputs, establish classroom norms for ethical use, and design activities that position AI as both a planning aid and a resource for student critique.

One of the most consistent insights from the co-design process was teachers' strong commitment to helping students read and write about complex, authentic science texts. They viewed the InSPECT and BDA frameworks as providing the structure needed to make this work manageable and effective within their classrooms. The teacher who implemented the lessons noted that her students responded positively to the organized, step-by-step approach, which supported their engagement and writing. At the same time, teachers emphasized that they wanted to connect with colleagues who shared this goal and have ample time to plan and create together. They valued opportunities to collaborate with others who were equally passionate about integrating literacy into science and requested extended work time within PD sessions to develop lessons, identify high-quality sources, and adapt materials for their students. Collectively, these insights informed a redesigned PD model that gives science teachers what they want and need: structured, collaborative, and adaptable support for helping students read, reason, and write about complex scientific ideas. In doing so, this study helps address a longstanding gap in high school science literacy instruction and provides a model with potential to improve students' engagement and comprehension of science from multiple texts.

4.3. Limitations and Future Directions

A limitation of this study is the relatively small sample size and the absence of direct measures of student learning outcomes. These features also reflect a strength of the design, as the co-design approach prioritized iterative refinement with teachers to ensure the PD was practical and responsive to classroom needs (e.g., [Cumbo & Selwyn, 2022](#); [Roschelle & Penuel, 2006](#)). Iteration and responsiveness to teacher feedback are essential steps in developing professional development that can be sustained in practice before scaling to larger contexts (e.g., [Blake & Traga Philippakos, 2025](#)).

Future research should extend this work by examining the long-term implementation and impact of the PD across schools and grade levels. Larger-scale studies that integrate teacher professional learning with systematic observation of classroom practice and student outcomes will provide critical evidence of effectiveness. Such efforts can build on prior research demonstrating the value of long-term, practice-based PD for shifting disciplinary literacy instruction (e.g., [Lupo et al., 2023](#); [Stevens et al., 2022](#); [Thomas & Drew, 2021](#)). Building on this foundation, future iterations of the project will continue to refine the

InSPECT and BDA frameworks and investigate how sustained PD cycles, embedded in authentic school contexts, can support both teacher learning and student achievement.

5. Conclusions

This study demonstrates the value of engaging teachers as co-design partners in refining professional development for disciplinary literacy with a specific emphasis on multiple-document reading and writing. By centering teachers' perspectives, the project produced concrete revisions to the PD model, including additional exemplars, collaborative planning opportunities, targeted attention to higher-order strategies for integrating across sources, and modules on generative AI. These refinements highlight how participatory design can bridge the gap between research and practice, resulting in PD that is both theoretically grounded and practically feasible. As a result, this work contributes to ongoing efforts to design sustainable professional learning that equips science teachers to support students in reading, evaluating, and writing from multiple documents in ways that improve engagement and learning.

Supplementary Materials: The following supporting information can be downloaded at <https://www.mdpi.com/article/10.3390/educsci16010006/s1>, Supplementary S1: Excerpt from Student Reading Guide 1; Supplementary S2: Excerpt from Student Reading Guide 4; Supplementary S3: Excerpt from Student Final Essay.

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Abbreviations

The following abbreviations are used in this manuscript:

PD	Professional development
InSPECT	Investigate, Source evaluate, Paraphrase, Explain, Connect, Transform ideas into writing
BDA	Before–During–After

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